

SIC-POVMs, a Stark Conjecture, and the 12th: A Formalization via Paraconsistent Belnap Multilattices

C. Lando Mills

Independent Researcher, Trabuco Canyon, CA 92679, USA

Correspondence: c.landonmills@gmail.com

ORCID: 0000-0003-0003-0552

Declarations

Availability of data and material

All Lean 4 modules cited in the manuscript live in the `p4ramill` library of the `p4rakernel` repository (<https://github.com/umpolungfish/p4rakernel>). Frozen review snapshot: branch `crystalline/manuscripts3-2026-07-07`, tag `crystalline-manuscripts` (commit `eea2c0c`). This manuscript and its companions `witness_vessel` and `chrysopoeia_2048` are frozen in `ig-docs` (<https://github.com/umpolungfish/ig-docs>) on the same branch name and tag.

Competing interests

The author declares no competing interests.

Funding

This research received no external funding.

Authors' contributions

C. Lando Mills: conceptualization, formalization, investigation, validation, writing (original draft), writing (review and editing).

Acknowledgements

The author thanks Harry T. Larson for imparting the importance of catching rising problems and never letting them go. The present formalization is written

in that spirit: the Stark–Zauner–Hilbert identification was not abandoned at the honest gap markers, and dimension $d = 12$ was carried from arithmetic prediction to a machine-checked existence theorem rather than left as a conjectural witness.